A Markov-Chain Approach to Citation-Based Paper Recommendation

Hayden Leung (cl4627), Rea Lila (rl3512), Mireille Donoso-Kugler (md4418)
Columbia University

***Abstract*—We model academic paper recommendation as Personalized PageRank on a citation network built around a single seed paper, scraped two hops outward via breadth-first search. The ranking we want comes out as the invariant distribution of a Markov chain whose walker, at each step, either follows a citation or teleports back to the seed; dangling nodes (papers with no recorded outgoing citations) are also redirected to the seed. To compute it, we solve a sparse linear system derived from this chain. We then test two predictions about the method's behavior. First, power iteration converges geometrically at a rate set by the damping factor times the magnitude of the second-largest eigenvalue of the transition matrix, with the observed contraction matching the spectral prediction within one percent. Second, the resulting ranking barely moves as we vary the damping factor across the range we tested: pairwise Pearson correlation between score vectors stays above 0.994 for every pair of damping values. On the seed *Attention Is All You Need*, the method produces a reading list that overlaps with raw citation count on 18 of 20 papers, with the remainder reflecting structural relevance beyond popularity.**

I. Introduction

Literature review for a new topic has a cold-start problem. A researcher comes across one interesting paper and wants to understand its intellectual neighborhood: the prior work it builds on, the follow-ups it has sparked, and the papers it should be read alongside. Keyword search is a blunt instrument for this because the right neighbors often share no surface terms. Citation count is a more structural alternative, but it conflates *popular in general* with *relevant to this specific paper*. The most cited paper in a field is almost never the most relevant to any given seed. The question we ask in this paper is whether a ranking that takes the seed into account can do better.

We approach this with PageRank, an algorithm originally developed for ranking web pages [1]. PageRank imagines a random walker who, at each step, either follows a link out of the current node (with some probability d called the *damping factor*) or, with probability $1 - d$, jumps to some chosen distribution of nodes. The long-run fraction of time the walker spends at each node is its rank. Mathematically, this fraction is the *invariant distribution* of a *Markov chain*, a random process whose next state depends only on the current state, summarized by a transition matrix between states. For a left-stochastic transition matrix, the invariant distribution coincides with the dominant eigenvector, so the Markov-chain view and the linear-algebra view of the course produce the same ranking. The *personalized* variant [2] fixes the walker's jump destination to a single chosen *seed* paper, biasing the resulting ranking toward papers structurally close to the seed rather than toward globally popular ones. We run the algorithm on the seed's *ego-network*, the subgraph of citations reachable within a small number of hops from the seed.

This paper does not propose a new algorithm. The personalized variant comes from Jeh and Widom [2], and the convention for handling papers with no recorded outgoing references comes from Langville and Meyer [3]. Gleich [4] surveys related applications, including bibliometric ranking on citation networks, and Kanakia et al. [5] build a hybrid recommender for Microsoft Academic that combines content and graph signals. What we contribute, then, is expository and empirical: we frame the algorithm in terms of the linear-algebra and probability content of the course, and we test its predictions on a real citation graph.

The empirical claim we make is narrow. On an ego-network scraped around a chosen seed, Personalized PageRank converges geometrically at a rate determined by the second-largest eigenvalue of the transition matrix, and the induced ranking holds steady against the damping factor across the range of values the literature considers reasonable. The remainder of the paper develops this claim in four steps. Section II sets up the mathematics and derives the linear system that Personalized PageRank solves. Section III analyzes the algorithm as a power iteration and checks the predicted convergence rate against the eigenvalues of P. Section IV measures the damping sensitivity of the resulting ranking. Section V compares the output to a raw-citation-count baseline and names the limitations.

II. Mathematical Setup

We model the scraped citation network as a directed graph on N papers and encode it in an $N \times N$ sparse adjacency matrix A, with $A_{ij} = 1$ if paper j cites paper i and zero otherwise. The column orientation is deliberate. A_{ij} records an edge *into* node i from node j, so each column of A enumerates the outgoing references of a single paper.

To turn this graph into a random walk, we divide each nonempty column of A by its column sum, and that single step gives us the *transition matrix*

$$P_{ij} = \frac{A_{ij}}{\sum_k A_{kj}}, \tag{1}$$

whose entry P_{ij} is the probability that a walker currently at paper j moves to paper i in one step. The walker picks one of paper j's references uniformly at random and follows it. We call P *left-stochastic* when every column sums to 1, that

is, when each column is a valid probability distribution over destinations.

The normalization fails on one class of columns. Papers with no outgoing edges in the scraped network (which we call *dangling nodes*) have a column of A that sums to zero, so the corresponding column of P is all zeros, which is not a probability distribution at all. These columns must be patched before P is properly left-stochastic.

We follow the *strongly preferential* convention of Langville and Meyer [3] and set every dangling column of P equal to the personalization vector $\mathbf{e}_s$, defined as the column vector whose entry at the seed paper is one and whose other entries are zero. Operationally, a walker that lands on a paper with no recorded references teleports back to the seed and continues from there. This keeps P left-stochastic, and it preserves the random-walk interpretation of the eventual invariant distribution [4].

With P now left-stochastic, we define the teleport-modified transition matrix

$$M = dP + (1-d)\,\mathbf{e}_s\mathbf{1}^\top, \tag{2}$$

where $d \in (0,1)$ is the damping factor and $\mathbf{1}$ is the column vector of ones. The outer product $\mathbf{e}_s\mathbf{1}^\top$ is the $N \times N$ rank-one matrix whose every column equals $\mathbf{e}_s$, so adding it blends a teleport-to-seed contribution into every column of M. At every step the walker either advances along the graph with probability d or teleports back to the seed with probability $1-d$.

Because each column of M is a convex combination of two stochastic vectors, M is left-stochastic and therefore a valid Markov transition matrix. Its principal eigenvector, with eigenvalue 1, is the invariant distribution $\mathbf{v}$:

$$M\mathbf{v} = \mathbf{v}. \tag{3}$$

This $\mathbf{v}$ is the *Personalized PageRank* vector, in the sense introduced by Jeh and Widom [2].

Because $\mathbf{v}$ is a probability distribution, $\mathbf{1}^\top\mathbf{v} = 1$, which collapses the rank-one term in M under multiplication by $\mathbf{v}$: $\mathbf{e}_s\mathbf{1}^\top\mathbf{v} = \mathbf{e}_s$. We substitute the definition of M into the eigenvector equation $M\mathbf{v} = \mathbf{v}$, distribute the multiplication, apply $\mathbf{e}_s\mathbf{1}^\top\mathbf{v} = \mathbf{e}_s$, and move $dP\mathbf{v}$ to the right-hand side:

$$\left(dP + (1-d)\,\mathbf{e}_s\mathbf{1}^\top\right)\mathbf{v} = \mathbf{v}, \tag{4}$$

$$dP\mathbf{v} + (1-d)\,\mathbf{e}_s\mathbf{1}^\top\mathbf{v} = \mathbf{v}, \tag{5}$$

$$dP\mathbf{v} + (1-d)\,\mathbf{e}_s = \mathbf{v}, \tag{6}$$

$$(1-d)\,\mathbf{e}_s = \mathbf{v} - dP\mathbf{v}, \tag{7}$$

$$(I - dP)\,\mathbf{v} = (1-d)\,\mathbf{e}_s. \tag{8}$$

Equation (8) is the linear system we solve in practice. This form has two advantages over the eigenvector formulation. First, $I - dP$ inherits the sparsity of P, so a sparse direct solver runs in near-linear time rather than computing a full eigendecomposition. Second, $I - dP$ is invertible, so the solution is unique. To see this, take any column j of $I - dP$. The diagonal entry is $1 - dP_{jj}$, and the absolute sum of the off-diagonal entries is $\sum_{i\neq j} d\,P_{ij} = d(1-P_{jj})$ by left-stochasticity. Whenever $d < 1$, the diagonal strictly exceeds this sum: $(1-dP_{jj}) - (d - dP_{jj}) = 1 - d > 0$. This property is called strict column diagonal dominance, and any matrix with it is invertible [3].

The solution $\mathbf{v}$ also sums to 1 on its own, so no post-normalization is needed. Left-multiplying (8) by $\mathbf{1}^\top$ and using $\mathbf{1}^\top P = \mathbf{1}^\top$ (left-stochasticity, again) gives $(1-d)\,\mathbf{1}^\top\mathbf{v} = (1-d)$, hence $\mathbf{1}^\top\mathbf{v} = 1$.

III. Method A: Power Iteration as a Recursive Sequence

Equation (8) is a shortcut to computing $\mathbf{v}$. But let's investigate what happens when we run the Markov chain step by step. Power iteration does exactly that, applying the transition rule once per iteration, and the rate at which the iterates converge to $\mathbf{v}$ is what we set out to predict in this section:

$$\mathbf{v}_{k+1} = dP\mathbf{v}_k + (1-d)\,\mathbf{e}_s, \tag{9}$$

starting from $\mathbf{v}_0 = \mathbf{e}_s$. At each step we push the current estimate one move along the graph (the $dP\mathbf{v}_k$ term) and add the constant teleport contribution back to the seed (the $(1-d)\mathbf{e}_s$ term). After enough iterations, $\mathbf{v}_k$ stops changing and equals $\mathbf{v}$. This is a recursive sequence of the kind studied in Nicholson [6].

Unrolling (9) k times gives the closed form

$$\mathbf{v}_k = (dP)^k\mathbf{e}_s + (1-d)\sum_{i=0}^{k-1}(dP)^i\mathbf{e}_s. \tag{10}$$

Two facts about the eigenvalues of P control how each term behaves as k grows. First, P is left-stochastic, so $\mathbf{1}^\top P = \mathbf{1}^\top$. The all-ones row vector is therefore a left eigenvector of P with eigenvalue 1, so 1 is an eigenvalue of P. Second, every eigenvalue of a left-stochastic matrix is bounded in modulus by 1 [3]: $|\lambda| \leq 1$ for every λ.

The largest absolute value across a matrix's eigenvalues is called the *spectral radius*, written $\rho(\cdot)$; Nicholson [6] calls this the *dominant eigenvalue*. The spectral radius controls the asymptotic rate at which the powers of a matrix grow or shrink. When $\rho(A) < 1$, $A^k \to 0$ as $k \to \infty$, and the smaller $\rho(A)$, the faster that decay. Combining the two facts above, $\rho(P) = 1$. Scaling P by d scales every eigenvalue by d (if $Pv = \lambda v$, then $(dP)v = (d\lambda)v$), so $\rho(dP) = d < 1$.

Both terms of (10) now have a clear fate. The first term $(dP)^k\mathbf{e}_s$ shrinks geometrically to zero because $\rho(dP) < 1$. The partial sum is a matrix geometric series. The telescoping identity $(I - dP)\sum_{i=0}^{n}(dP)^i = I - (dP)^{n+1}$ tends to I as $n \to \infty$ since $\rho(dP) < 1$, so the limit is the Neumann series $\sum_{i=0}^{\infty}(dP)^i = (I - dP)^{-1}$, the matrix analogue of $\sum_i d^i = 1/(1-d)$ (Nicholson [6] derives this expansion without naming it). Substituting these limits into (10) gives

$$\mathbf{v}_\infty = (1-d)(I - dP)^{-1}\mathbf{e}_s, \tag{11}$$

which is exactly the solution of (8). Power iteration and the linear solver produce the same $\mathbf{v}$.

We run the recursion at the conventional damping $d = 0.85$ on the ego-network scraped from the seed *Attention Is All You*

Need [7] on 2026-04-25, which contains $N = 2{,}504$ papers, and track the residual $r_k = \|\mathbf{v}_k - \mathbf{v}_{k-1}\|_1$ at each iteration (the $\|\cdot\|_1$ norm sums the absolute values of the entries of a vector).

How fast should r_k decay? Subtracting two consecutive iterates of (9) cancels the constant teleport term and leaves

$$\mathbf{v}_{k+1} - \mathbf{v}_k = dP(\mathbf{v}_k - \mathbf{v}_{k-1}),$$

so successive differences themselves evolve under the linear map dP. Each such difference has entries that sum to zero, because $\mathbf{1}^\top \mathbf{v}_k = \mathbf{1}^\top \mathbf{v}_{k-1} = 1$ for any two probability distributions. The differences therefore live in the subspace of zero-sum vectors, and this subspace is invariant under dP (a zero-sum vector multiplied by dP stays zero-sum, since $\mathbf{1}^\top dP = d\mathbf{1}^\top$). The right eigenvector of P for $\lambda_1 = 1$ is an invariant distribution of P, with nonnegative entries summing to one. Since the entries sum to one rather than zero, it does not lie in the zero-sum subspace. Restricted to that subspace, then, dP acts only through eigenvalues other than λ_1. The rate at which dP shrinks differences is therefore set by the next-largest eigenvalue of P in magnitude, $\lambda_2(P)$. The residuals decay at rate $d \cdot |\lambda_2(P)| \le d$, with strict inequality whenever there is a gap between $|\lambda_1| = 1$ and $|\lambda_2|$, called the *spectral gap* of P [3].

Figure 1 plots the result. Power iteration drives the residual below 10^{-10} in 74 iterations, with each step shrinking the residual by roughly 0.729, comfortably below the loose bound $d = 0.85$. Computing all eigenvalues of P directly with a sparse eigensolver gives $\lambda_1 = 1$ as required and a complex conjugate pair $\lambda_2 = -0.418 \pm 0.741i$. Complex eigenvalues of a real matrix come in conjugate pairs, and the contraction rate depends on their magnitude $|\lambda| = \sqrt{a^2 + b^2}$, here $\sqrt{(-0.418)^2 + (0.741)^2} = 0.851$. The spectral-gap argument above then predicts a per-step contraction of $d \cdot |\lambda_2(P)| \approx 0.723$, matching the observed 0.729 within 1%. The remaining eigenvalues are an order of magnitude smaller ($|\lambda_3| \approx 0.07$), so the scraped graph has a clean spectral gap and power iteration converges as predicted by a two-eigenvalue model.

IV. Method B: Damping Sensitivity

Section III fixed $d = 0.85$ throughout. The question now is whether that choice matters: does the ranking shift much if we slide d across the range the literature considers reasonable? The damping factor carries two meanings simultaneously. As a Markov chain parameter, d is the probability that the walker continues along the graph rather than teleporting back to the seed. As a linear-algebra parameter, d is the *contraction factor*. The recursion shrinks the gap between successive iterates by roughly d each step, which is what makes them converge. These two roles connect through the teleport itself. Every time the walker teleports back to $\mathbf{e}_s$, the non-invariant components of the current iterate shrink by a factor of d. A smaller d teleports more often, converges faster, and biases scores toward the seed; a larger d explores more of the graph but converges more slowly.

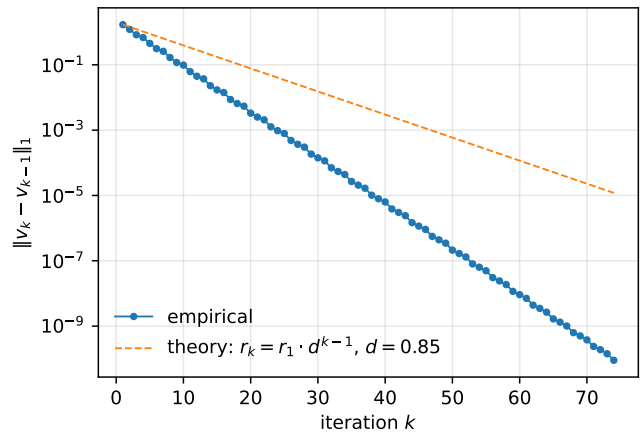

Fig. 1. Convergence of power iteration on the *Attention Is All You Need* ego-network ($N = 2{,}504$, $d = 0.85$). The residual $\|\mathbf{v}_k - \mathbf{v}_{k-1}\|_1$ shrinks by a factor of ≈ 0.729 per step, faster than the loose bound $d = 0.85$ (dashed). Computing $|\lambda_2(P)| = 0.851$ directly gives the spectral-gap prediction $d \cdot |\lambda_2(P)| \approx 0.723$, which matches the observed rate within 1%.

We used $d = 0.85$ in Section III because it is the value Brin and Page originally chose [1] and the value the PageRank literature has retained ever since. The choice is conventional rather than derived. A sensible ranking method should not depend strongly on the exact choice of its parameters, so we quantify how much the ranking changes as we vary d across the range $[0.50, 0.99]$.

We compute the Personalized PageRank vector $\mathbf{v}_d$ at each $d \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.99\}$ and record the pairwise Pearson correlation

$$r(\mathbf{v}_{d_1}, \mathbf{v}_{d_2}) = \frac{\mathrm{Cov}(\mathbf{v}_{d_1}, \mathbf{v}_{d_2})}{\sigma_{d_1}\, \sigma_{d_2}}, \tag{12}$$

treating each paper as an observation. This is the correlation coefficient studied in the covariance-and-correlation chapter of the course [6]. A value close to 1 indicates that the two score vectors agree up to a linear transformation, a sufficient condition for them to induce the same ranking.

Figure 2 shows what we get. The lowest off-diagonal correlation across the entire sweep is $r(\mathbf{v}_{0.50}, \mathbf{v}_{0.99}) = 0.994$; within the literature-recommended band $d \in [0.70, 0.95]$, every pair exceeds 0.998. For practical purposes the ranking does not depend on d on this graph at all. Any reasonable choice of damping produces the same reading list.

Two things drive this stability, and they reinforce each other. First, the top of the ranking sits well clear of the rest in absolute score. The seed itself accounts for ≈ 0.389 of the total score and the second-ranked paper for only ≈ 0.024, so small relative perturbations cannot reorder the top ranks. Second, d enters $\mathbf{v}$ only through how strongly each eigenvector of P contributes, not through the eigenvectors themselves. The dominant structure of $\mathbf{v}$ is set by the leading eigenvectors of P regardless of damping.

This is a claim about ranks, not absolute scores. Pearson correlation measures whether two vectors agree up to a linear

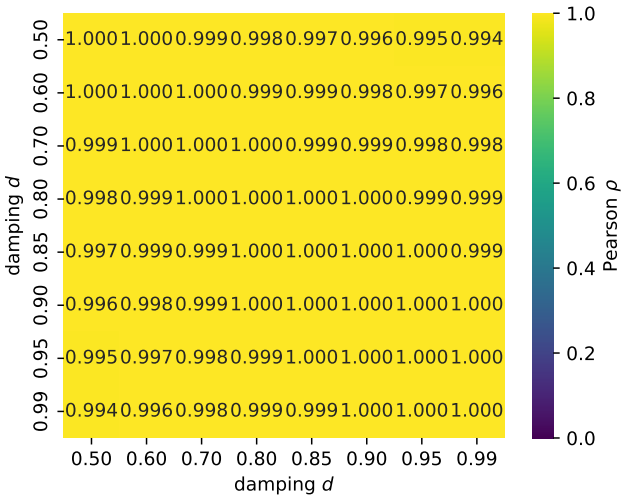

Fig. 2. Pairwise Pearson correlation of PPR score vectors across damping values on the *Attention Is All You Need* ego-network. Every off-diagonal entry exceeds 0.994; within the literature-recommended band $d \in [0.70, 0.95]$, every pair exceeds 0.998.

transformation, which captures the *ordering* of the entries but is insensitive to small shifts in the score values themselves. The distinction matters here: as $d \to 1$, the smallest eigenvalue of $I - dP$ is $1 - d$ (corresponding to $\lambda_1 = 1$ of P), which approaches zero. Inverting $I - dP$ then amplifies perturbations by roughly $1/(1-d)$, so small numerical changes to the graph or the right-hand side can produce large changes in the score values $\mathbf{v}$. A linear system with this property is called *ill-conditioned* [3]. The scores wobble; the ranks do not.

V. Discussion

With convergence and damping stability established, we now ask whether PPR's ranking actually differs from the simplest alternative.

The simplest alternative we can compare against is to rank papers by raw citation count, that is, by the *in-degree* of each node in the scraped graph (the number of other papers in our sample that cite it). Table I shows the two top-20 lists side by side. Within the top-10 they overlap on 7 of 10 papers; across the full top-20, on 18 of 20. Where they diverge is informative.

The most striking difference is at the very top. PPR ranks the seed itself, *Attention Is All You Need*, at #1, while the seed does not appear in the citation-count top-20 at all. This difference is the personalization at work. The teleport vector $\mathbf{e}_s$ sends probability back to the seed at every step, so the seed accumulates score regardless of how many scraped papers cite it. In-degree only sees citations inside our two-hop window, which captures a small fraction of the seed's total incoming edges.

Within the top-10 specifically, the PPR-only entries (Huang 2009, Kaiser 2016) are also direct references of the seed, as are the citation-count-only entries (Luong 2015, Wu 2016, Marcus 1993). The difference between the two lists is therefore not about graph distance: all five papers sit one citation hop from the seed.

TABLE I
Top-20 papers in the *Attention Is All You Need* ego-network ranked by Personalized PageRank versus by raw citation count (in-degree). Stars mark papers unique to each list. The seed sits at the top of the PPR ranking but does not appear in the citation-count top-20 at all.

Rank	PPR ranking	Citation-count ranking
1	Vaswani 2017 (seed)⋆	Hochreiter 1997
2	Huang 2009	Bahdanau 2014
3	Kaiser 2016⋆	Sutskever 2014
4	Hochreiter 1997	Cho 2014
5	Sutskever 2014	Graves 2013
6	Bahdanau 2014	Kingma 2014
7	Cho 2014	He 2015
8	Graves 2013	Luong 2015
9	He 2015	Wu 2016
10	Kingma 2014	Marcus 1993
11	Marcus 1993	Zhou 2016
12	Luong 2015	Hochreiter 2001
13	McClosky 2006	Vinyals 2014
14	Wu 2016	Sennrich 2015
15	Petrov 2006	Sukhbaatar 2015⋆
16	Vinyals 2014	McClosky 2006
17	Hochreiter 2001	Józefowicz 2016
18	Zhou 2016	Huang 2009
19	Józefowicz 2016	Ba 2016⋆
20	Sennrich 2015	Petrov 2006

What separates them, then, is how each ranking weights incoming edges. Across the network as a whole the two scores agree strongly: the Pearson correlation between PPR and in-degree (over all 2,503 non-seed papers) comes out at 0.958, so the rankings are mostly aligned. The reorderings in Table I are where the structural-versus-popular distinction surfaces. Citation count weighs every citer equally and tracks raw in-degree: Luong, Wu, and Marcus carry 59, 57, and 57 incoming edges in our scrape, above Huang's 53 and Kaiser's 52. PPR weighs each incoming edge by the score of the citer, and the mean PPR score among the citers of Huang and Kaiser is roughly twice that of the citation-count-only entries (≈ 0.015 versus ≈ 0.008). Huang and Kaiser are therefore embedded more deeply in the seed's neighborhood, while Luong, Wu, and Marcus are cited more often but by lower-ranked papers.

Three limitations are worth flagging up front. First, the scrape itself was bounded. We explored outward from the seed, fetching at most 50 references and 50 citations per paper and stopping after two hops. This bounds graph size but also truncates papers with dense citation neighborhoods. Concretely, for the seed *Attention Is All You Need*, the cap captures all 41 outgoing references but only 50 of its 173,976 incoming citations on Semantic Scholar: full coverage of what the seed builds on, but a tiny sample of what builds on the seed. Second, the seed choice dominates the result. The PPR ranking on this graph reflects what is close to *Attention Is All You Need*, not what is globally important; running the pipeline with a different seed would scrape a different ego-

network and produce a different ranking. Third, the Semantic Scholar API [8] has real coverage gaps. In our run, 56 of the 89 papers reachable in one citation hop from the seed returned null reference data because the publisher had not released the metadata, and papers outside S2's coverage entirely cannot be processed at all.

VI. Conclusion

We modeled academic paper recommendation as Personalized PageRank on a directed citation ego-network, and we looked at the resulting system from two angles. Power iteration converges geometrically at rate $d \cdot |\lambda_2(P)|$, with the observed contraction (0.729) matching the spectral prediction (0.723) within 1%. The ranking barely budges under the damping parameter. Pairwise Pearson correlation across $d \in [0.50, 0.99]$ exceeds 0.994, so any reasonable choice of d produces the same reading list. Natural extensions include deeper scrapes, sparser seeds, and comparison against topic-model recommenders.

Our code, the canonical scrape, and the figures used in this paper are available at https://github.com/hayden1126/PaperRec-submission.

References

[1] S. Brin and L. Page, "The anatomy of a large-scale hypertextual Web search engine," *Computer Networks and ISDN Systems*, vol. 30, no. 1–7, pp. 107–117, 1998.

[2] G. Jeh and J. Widom, "Scaling personalized web search," in *Proceedings of the 12th International Conference on World Wide Web (WWW)*, 2003, pp. 271–279.

[3] A. N. Langville and C. D. Meyer, "Deeper inside PageRank," *Internet Mathematics*, vol. 1, no. 3, pp. 335–380, 2004.

[4] D. F. Gleich, "PageRank beyond the web," *SIAM Review*, vol. 57, no. 3, pp. 321–363, 2015.

[5] A. Kanakia, Z. Shen, D. Eide, and K. Wang, "A scalable hybrid research paper recommender system for Microsoft Academic," in *Proceedings of the World Wide Web Conference (WWW)*, 2019, pp. 2893–2899.

[6] W. K. Nicholson, *Linear Algebra with Applications (Open Edition, Version 2018A)*. Lyryx Learning, 2018.

[7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[8] Semantic Scholar, "Semantic Scholar Academic Graph API," https://api.semanticscholar.org/, accessed 2026.